

ate EV sales. As 75 per cent of Indian commuters commute only about 35 kilometres a day, any EV meeting this range requirement will, hence,

be sufficient to convert 75 per cent of vehicles onroad to EVs. Combine this with the fact that consumer vehicles are parked eighty to ninety per cent

of the day, this time can be used to slow charge these EVs at home, office, malls, etc.

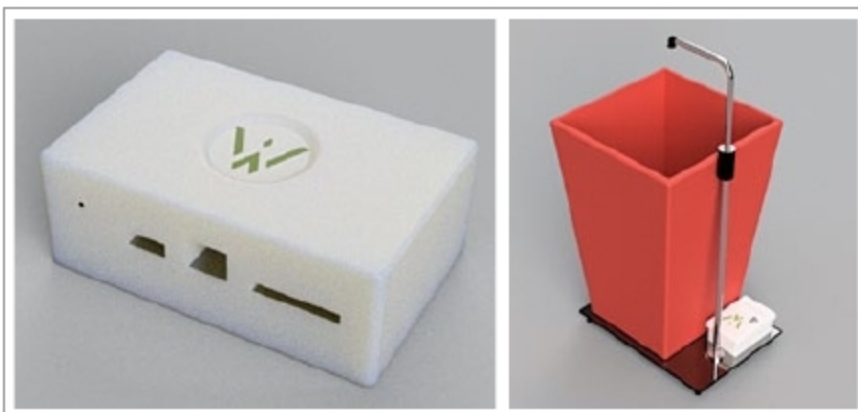
—Deepshikha Shukla

2 HOW THE IoT IS BEING UTILISED FOR A CLEANER INDIA

Rising population and urbanisation in India has led to a massive increase in waste generation. Ensuring better living conditions by providing a clean environment is the primary focus of any smart city. This requires efficient waste management for proper disposal and recovery, which presents a huge market opportunity. Founded in 2018, IIT-Madras-incubated startup Antariksh Waste Ventures (AirB.in) aims to digitise the waste management ecosystem by using the Internet of Things (IoT) to streamline the waste collection process.

Mahek Mahendra Shah, director, Antariksh Waste Ventures, says, “I created SwachhBharatApp.com in 2014. It has been adopted by the government via NGO funding as the official Swachata app. It has generated more than 3.5 million complaints from citizens in the last couple of years. An analysis of thousands of crowdsourced citizen complaints was done via the app, and our team identified garbage holding a potential business opportunity that could be catered to, using new smart city technologies in India.”

With the motto Clear Every Bin Before It Overflows in mind, the company designed and developed the IoT-powered smart garbage bin called AirBin (artificial intelligence radio bins) along with smartphone app for management. Sensors track garbage levels in the bin and on reaching



maximum capacity send signals to the IoT module.

Sanitation workers come under the unorganised sector in the country. AirBin app was released in October 2018 on Android to ease their daily tasks. The app notifies the workers to clear the garbage immediately. Administrators can also use the app for marking the attendance of staff and monitoring their daily work in real time.

The company improved prototype bins by working on PCB optimisation through size reduction and fibre-glass base fabrication. Fibre-glass was selected as the suitable material due to its properties like flame resistance, light weight and so on.

In 2019, the app was released for citizens also, to aid them in conveying issues in their locality and sync their waste disposal with pickup timings. To find AirBin in an area, users need to select the search icon on the home page in the app and click on map option. After finding the bin, they can mark it as a favourite.

The app empowers stakeholders

(residents, sanitation staff or municipal authorities) through engagement and interaction. This provides an opportunity for improved services. In addition, users have full control over their shared information, and data is only stored till it is necessary. Shah says, “Our system is low power NB-IoT and LoRa

(smart city) scalable technology, powered, designed, tested and being built for Indian conditions, geography and challenges.”

The solution caters to universities, industrial zones, townships and so on. Regarding pricing, Shah says, “Pricing structure depends on customer segment. For municipality, we will be proposing via NBFC financing models. Cost of smart bins will be higher than the current containers.”

The pilot project of the startup in Valmiki Nagar (Thiruvannamipur), Chennai, has been completed recently. The firm was also shortlisted for Qualcomm Design in India Challenge and received an amount of US\$ 10,000. Shah says, “Presently, we are aiming to get a smart city or municipal corporation to give us a paid work order in any city or area. After the launch of our industrial product, we are also targeting 2000 smart bins orders by 2020 end, provided that the government’s and municipalities’ focus on cleanliness and smart cities remains a priority.”

—Ayushee Sharma

3 AI-BASED INTELLIGENT BAGGAGE SCANNER SYSTEM

With millions of people traveling during the holidays, it is a busy time for any airport and/or metro station. Long lines,

delays and pushing carry-ons into a scanner are common things people encounter. Currently, information is sent to the computer it is connected

to, which identifies the objects using identification algorithms. Algorithms calculate z effective number (number of protons in a nucleus) and mass den-